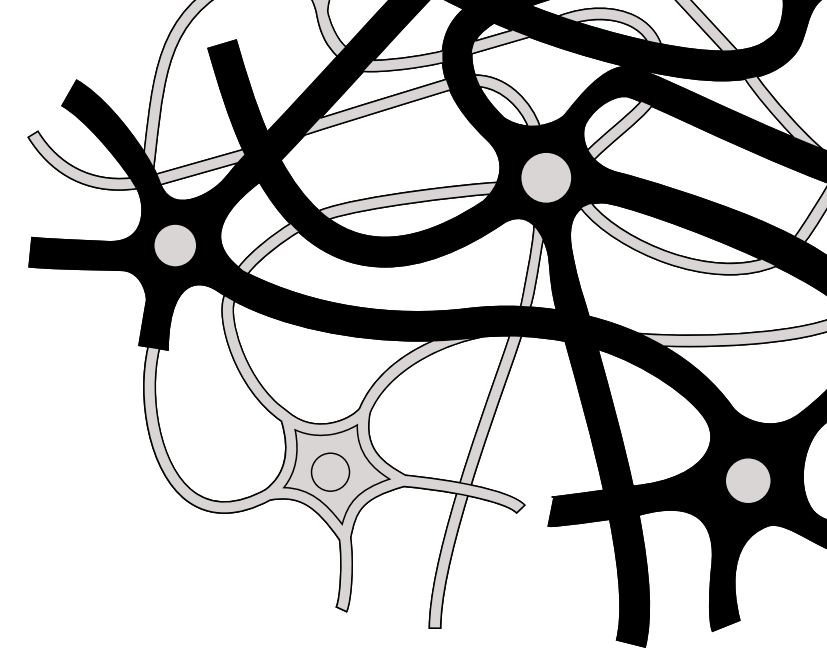
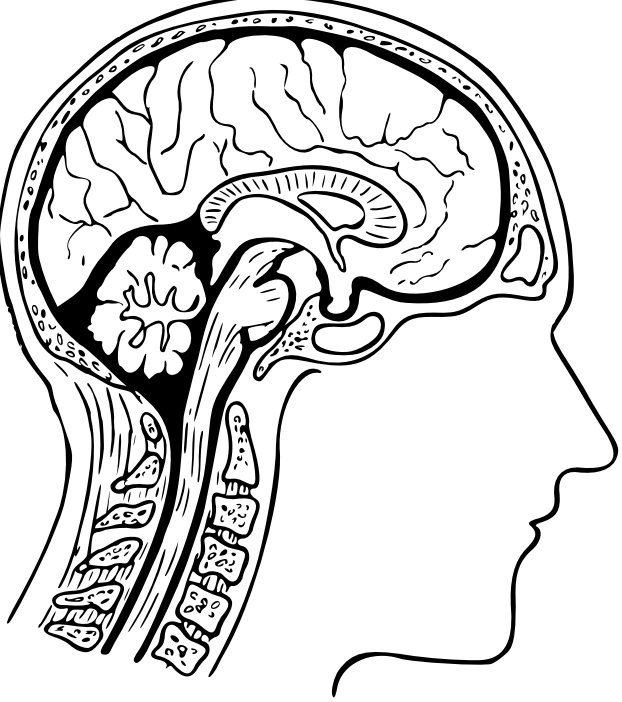
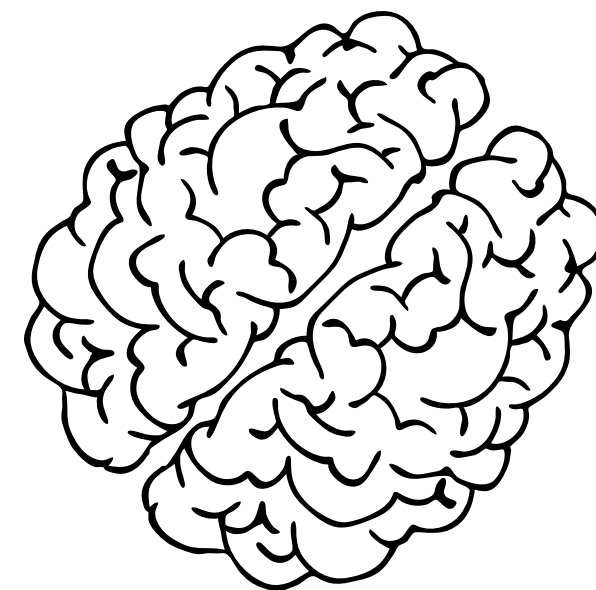
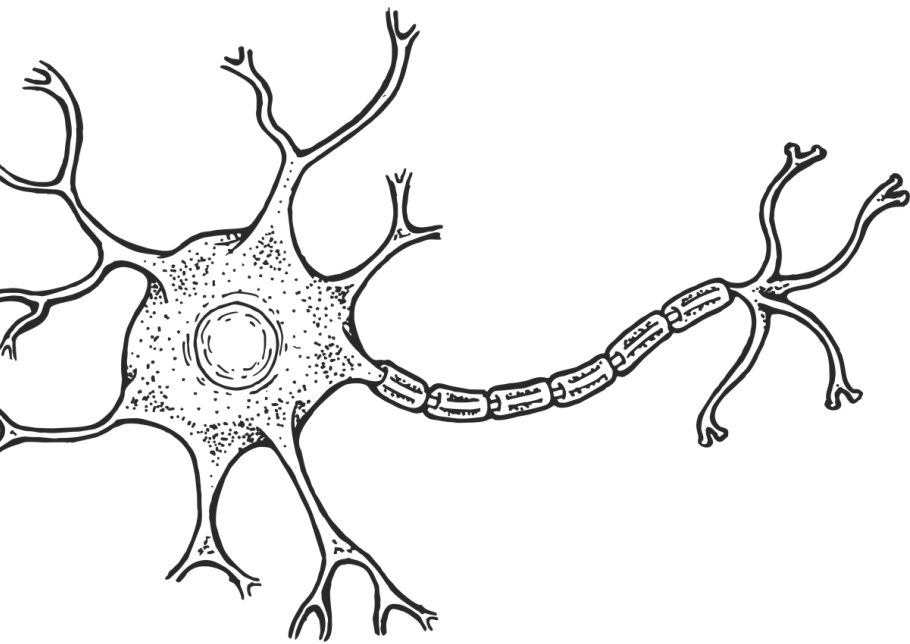
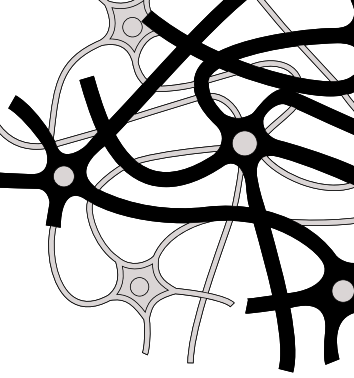
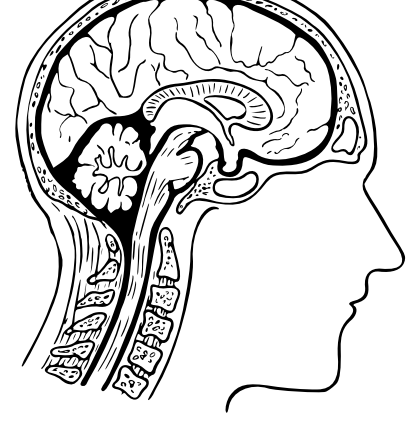




Session 19: Epoch automation and condition splitting

presented by Lynn





Learning Goals

- Segment continuous clinical data into discrete epochs.
- Implement baseline correction protocols.
- Apply threshold-based rejection algorithms.

Part A: Event Parsing

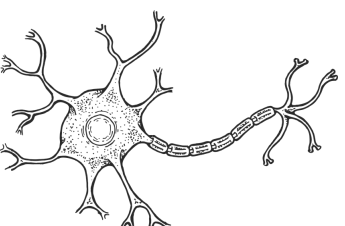
- Extracting triggers, defining windows (-200ms to 800ms), and applying baseline correction.

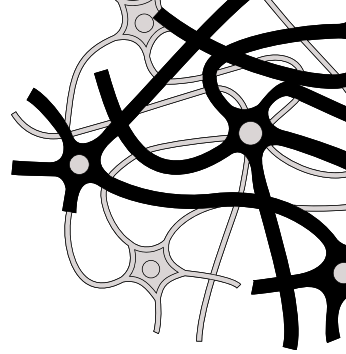
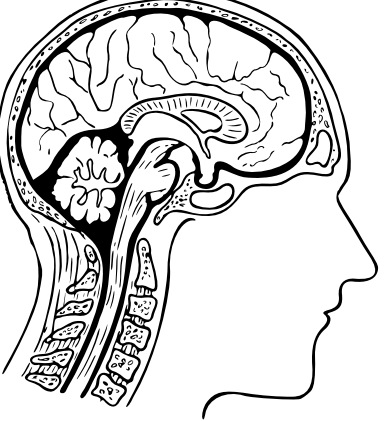
Part B: Threshold Rejection

- Replacing visual inspection with strict peak-to-peak amplitude parameters.

Part C: Condition Splitting

- Creating distinct arrays for analysis (e.g., Seizure vs. Non-Seizure).

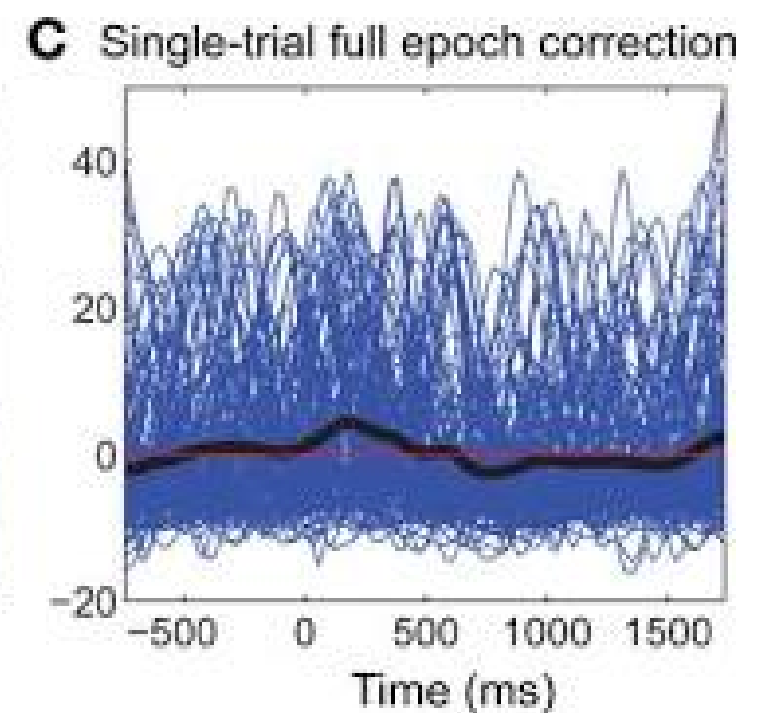
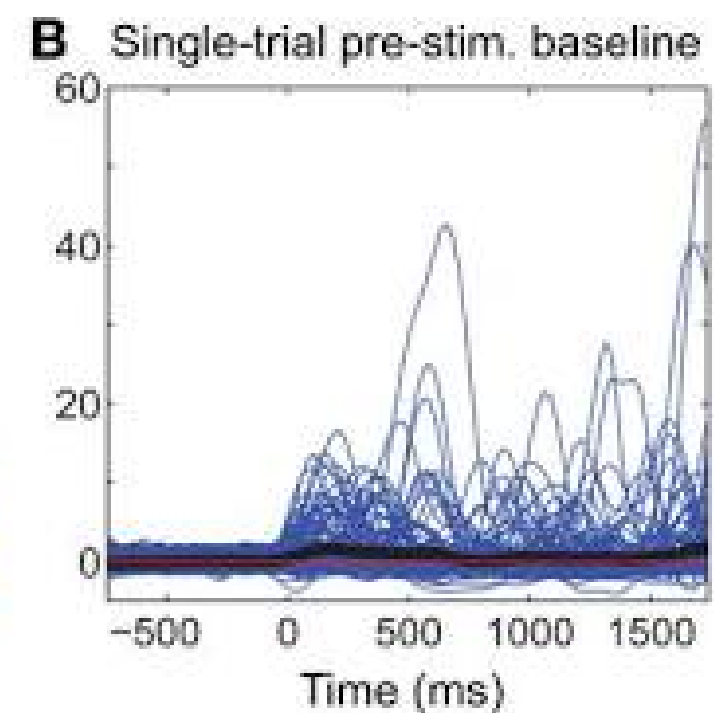
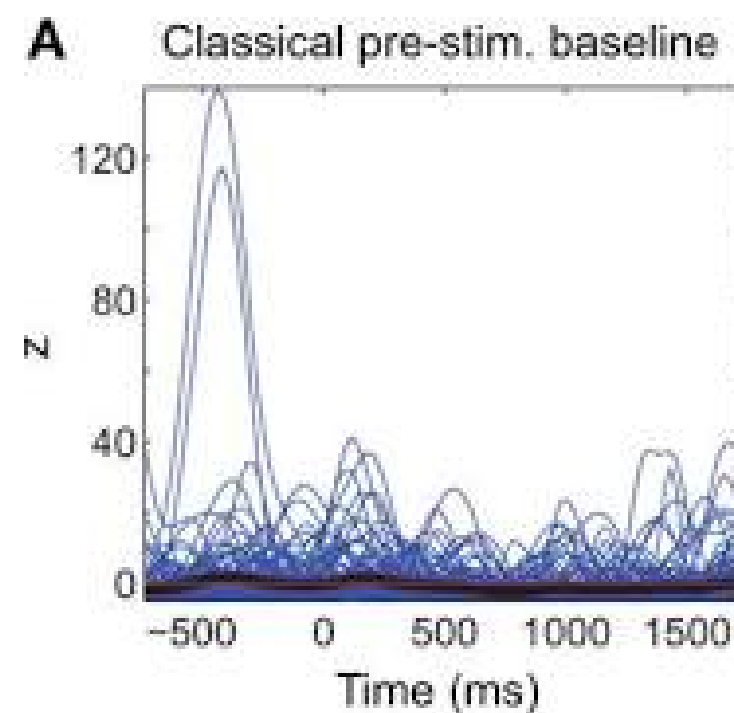




Event Parsing & Time Locking

From Continuous to Discrete

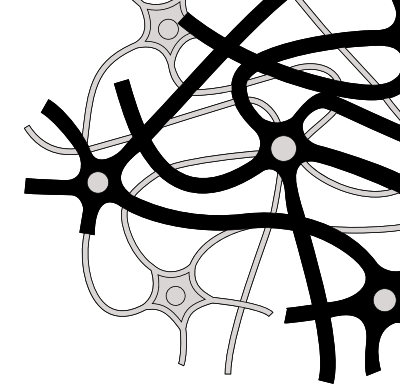
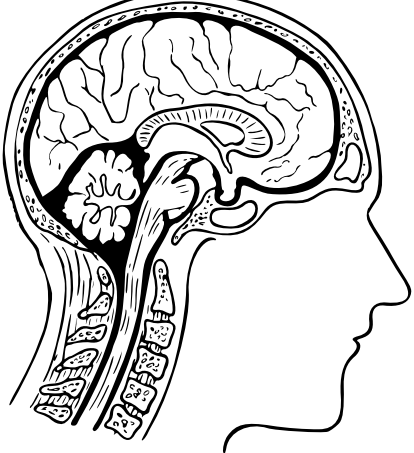
- We must slice continuous clinical EEG into time-locked epochs around specific trigger annotations.
- **Window of Interest:** Define the temporal window relative to the event. For many cognitive ERPs, this is $t_{min} = -0.2$ (200ms pre-stimulus) to $t_{max} = 0.8$ (800ms post).
- **Baseline Correction:** Essential for comparing voltages. MNE can automatically subtract the pre-stimulus average.



```
# Define epoch parameters
tmin, tmax = -0.2, 0.8
baseline = (None, 0)

# Create epochs object
epochs = mne.Epochs(
    raw, events, event_id, tmin, tmax,
    baseline=baseline, preload=True
)
```





Threshold-based Rejection

Manual visual inspection is subjective and time-consuming. We replace it with systematic algorithms.

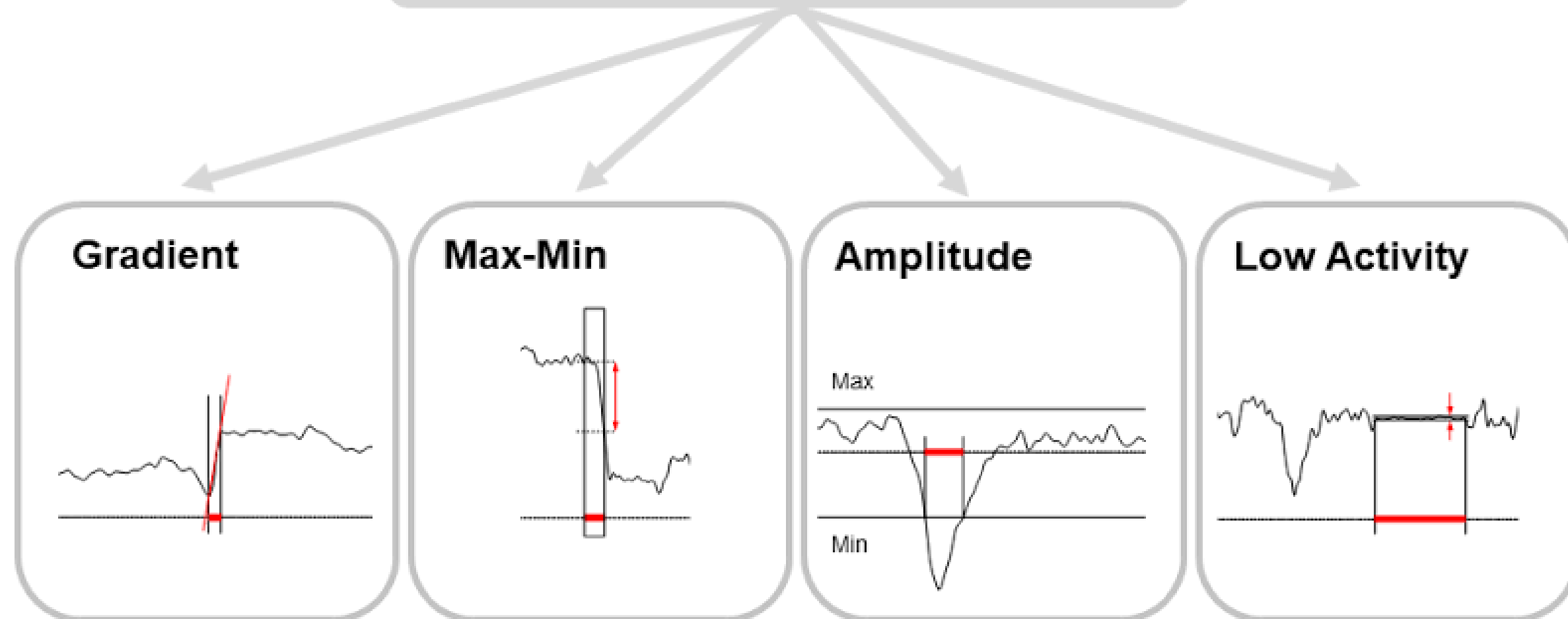
Peak-to-Peak Amplitude Rejection

- We instruct MNE to automatically drop any epoch where the signal fluctuation exceeds a strict physiological threshold (e.g., 100-150 μV), which typically indicates blinks, muscle movement, or electrode pops.

The reject Dictionary

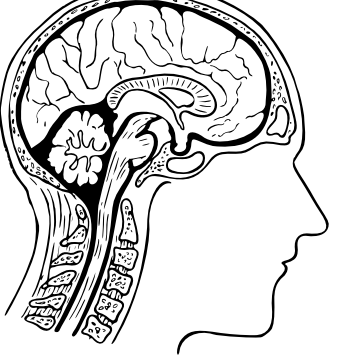
- Passed directly during Epoch creation to drop bad trials on the fly.

Automatic artifact rejection criteria

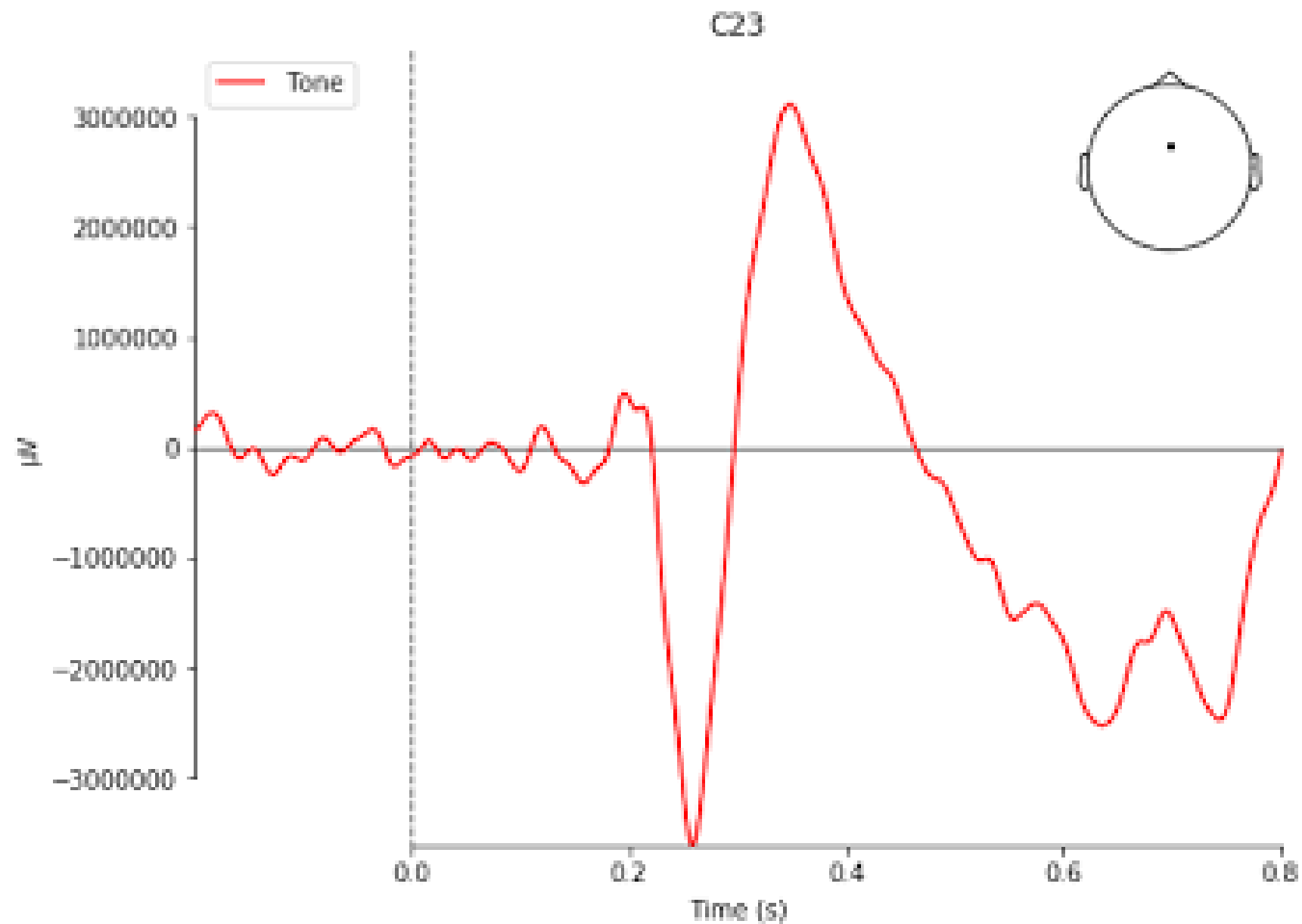
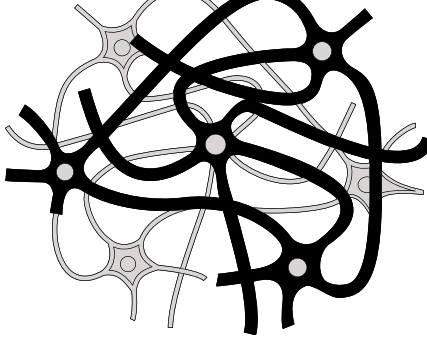


```
# Reject epochs exceeding 150  $\mu\text{V}$  (150e-6 Volts) on EEG channels
reject_criteria = dict(eeg=150e-6)

clean_epochs = mne.Epochs(raw, events, event_id, tmin, tmax, reject=reject_criteria)
```



Condition Splitting

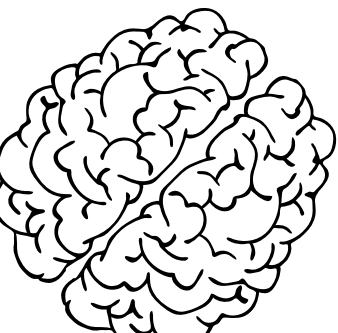


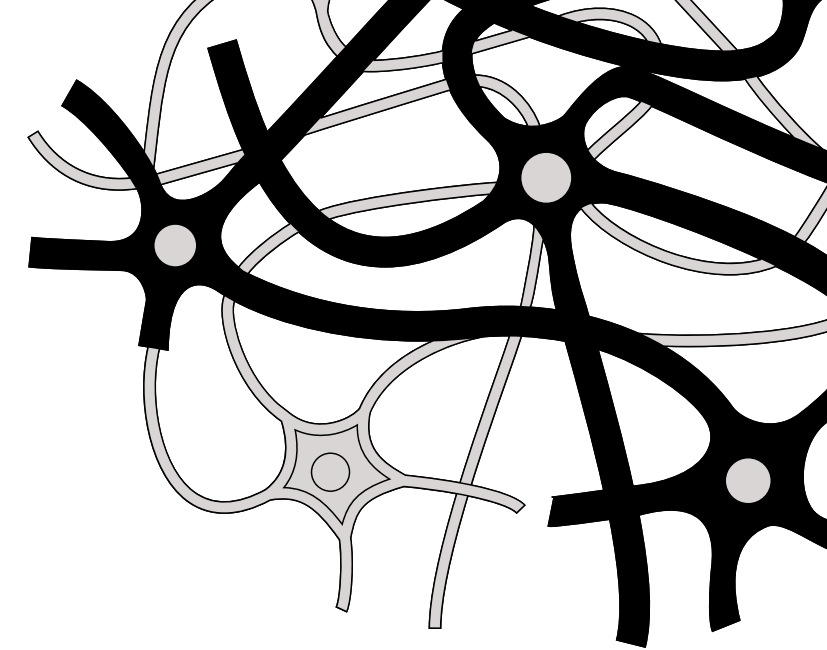
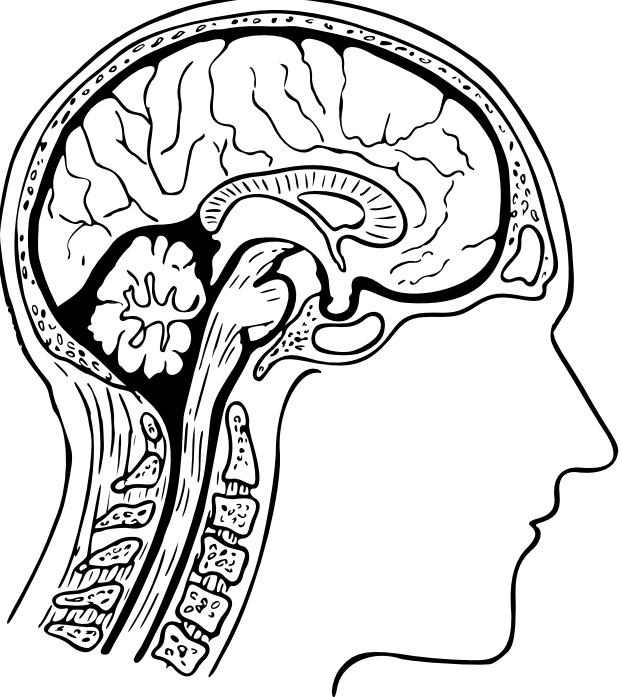
Once data is epoched and cleaned, we must isolate distinct conditions to run statistical comparisons.

- **Dictionary Access:** MNE allows you to split the Epochs object easily using dictionary-like string access based on your event_id.
- **Distinct Arrays:** We create separate objects (e.g., epochs['Seizure'] vs. epochs['Control']) to compute independent Evoked averages.

```
# Split the cleaned epochs by condition string
epochs_seizure = epochs['Seizure']
epochs_non_seizure = epochs['Non-Seizure']

# Average into Evoked objects for plotting/stats
evoked_sz = epochs_seizure.average()
evoked_nsz = epochs_non_seizure.average()
```





**Thank
You**

